

Recycling of Polyesters by Organocatalyzed Methanolysis Depolymerization: Environmental Sustainability Evaluated by Life Cycle Assessment

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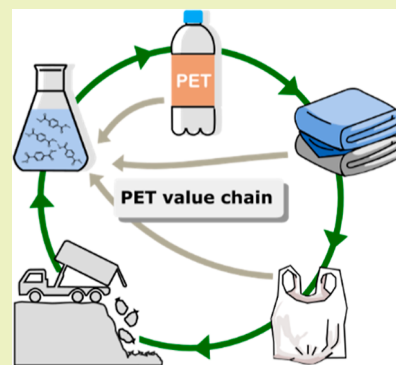
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ABSTRACT: Polyethylene terephthalate (PET) is one of the most common plastics and can be cascaded mechanically during its life cycle. However, recycling affects the mechanical properties of the material, and the virgin material is constantly in demand. If a worn material could be depolymerized to its chemical building blocks, then a virgin polymer could be generated from old fibers. In this work, we have developed a benign organo-catalytic depolymerization of PET to yield dimethyl terephthalate (DMT) and ethylene glycol (EG) without the need for purification of generated monomers. By recirculating the solvent and organo-catalyst, a solvent/substrate ratio of 3:1 was achieved. The depolymerization was successfully applied to other polyesters, polycarbonates, and polycotton. The cotton isolated from the polycotton depolymerization was successfully processed into viscose fibers with a tenacity in the range of nonwaste cotton-derived viscose filaments. The global warming potential (GWP) of PET depolymerization was evaluated by using life cycle assessment (LCA). The GWP of 1 kg PET recycling is 2.206 kg CO₂ equivalent, but the process produces DMT, EG, and heat, thereby avoiding the emissions equivalent to 4.075 kg CO₂ equivalent from the DMT, EG, and steam-energy production through conventional pathways. Thus, the net result potentially avoids the emission of 1.88 kg of CO₂ equivalent. The impact of this process is lower than that of waste PET incineration and conventional PET recycling technologies.

KEYWORDS: PET depolymerization, polycotton, textile recycling, environmental sustainability, life cycle assessment



INTRODUCTION

According to a Statista report, the global production of polyethylene terephthalate (PET) in 2020 was 73 million metric tons.¹ It has been forecasted that the global demand will increase with 42 million metric tons by 2030.² PET is one of the most commonly used thermoplastic polymers and finds applications in a broad spectrum of different sectors such as packaging, textile industry, medical equipment, construction of automobile, and electronic appliances.³ Even though PET is a polyester, it is robust and does not easily degrade. This has advantages in several applications but also disadvantages when it comes to plastic pollution where it does not degrade in nature. The life-cycle of PET can involve first a premium application such as a food container that can be recycled up to 5–6 times.⁴ After this, the fiber can be utilized in second-use consumer applications such as textile fibers or bags. Finally, the fiber will be incinerated or end up in landfills.^{5,6} As the demand of the virgin fiber for premium applications is steadily growing, recycling of the plastic fiber to its monomeric form and by this way generating a virgin fiber has gained attention.^{7,8} There are different routes to recycle PET to its building blocks^{9,10}

(Scheme 1): hydrolysis, glycolysis, and alcoholysis.^{11–13} Hydrolysis of PET to terephthalic acid (TA) and ethylene glycol (EG) can be performed in either neutral conditions or acid- or base-catalyzed conditions (Scheme 1a). All three methods have disadvantages. The noncatalyzed hydrolysis requires high temperatures (300 °C), the acid-catalyzed reactions can be performed at 100 °C, however the corrosiveness of equipment is a severe drawback and the base-catalyzed hydrolysis is performed at around 250 °C and usually employs an inorganic base that needs stoichiometric neutralization and further purification. Recently, rapid hydrolysis of PET in high-concentration alcohol using KOH as a catalyst was performed at 80 °C for 60 min, where more than 98% of TA was recovered by simple purification and using

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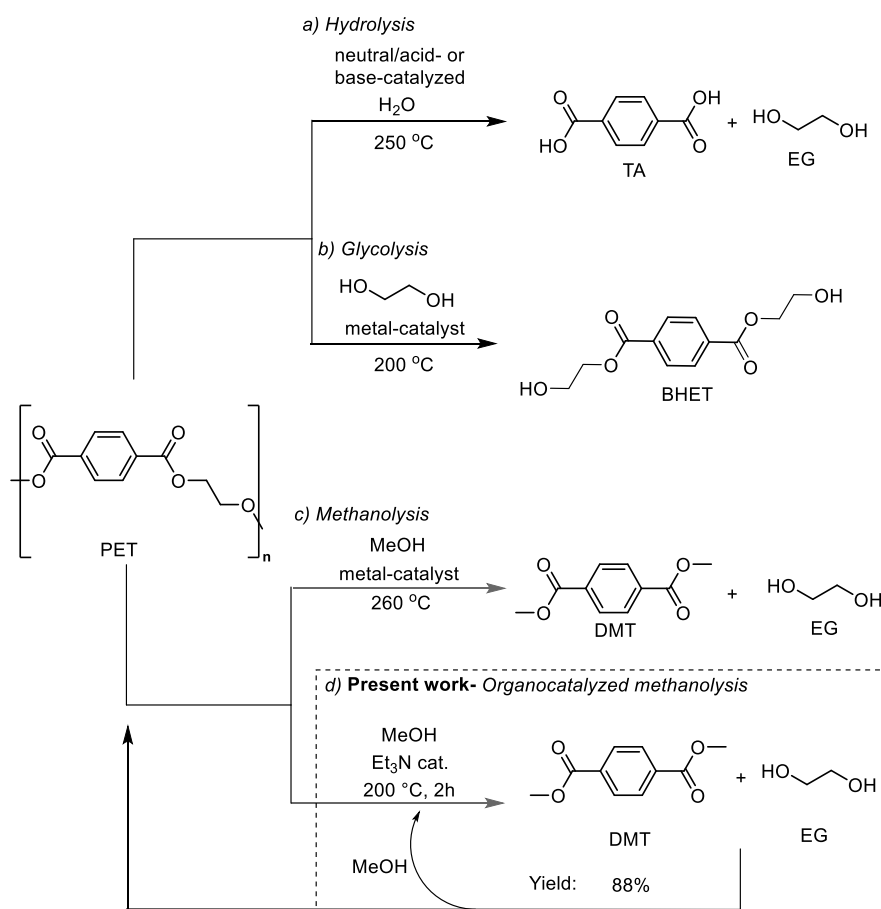
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Scheme 1. Depolymerization Reactions of PET Fibers Into DMT and EG



small amounts of acid during neutralization.¹⁴ However, a 5:1 base to substrate ratio was used, and recovery of EG was not reported. The most employed PET recycling methodology is glycolysis to yield the ester of EG and TA: Bis-2-hydroxyethyl terephthalate (BHET) (Scheme 1b).^{15–18} The depolymerizations are usually employed using a metal catalyst at $250\text{ }^\circ\text{C}$. Alcoholysis uses an alcohol such as MeOH to produce dimethyl terephthalate (DMT) and EG.^{19–22} (Scheme 1c) The reactions are usually performed at $250\text{ }^\circ\text{C}$ and require the use of a metal catalyst, which in turn entails an additional step for separating the solid catalyst from the solid DMT. These complex product mixtures make the separation and refinement very tedious and costly.^{7,19,23} Recently, an organocatalytic methanolysis reaction of PET using ammonium salt was reported that was performed at $100\text{ }^\circ\text{C}$ for 16 h using solvent to PET ratios of 20:1 to give DMT in 72% yield.²⁴ The reactions were performed in a glovebox under argon atmosphere. Purification of DMT by washing steps was required, probably due to the ammonium salt used as catalyst and the formation of byproducts. In recent years, many studies focused on developing solvent systems comprising ionic liquids and deep eutectic for the depolymerization of polyesters.^{25–30} However, purification is required by acidic neutralization to precipitate TA, where the filtrate contains a mixture of EG and ionic liquid solvents. Here, we report an organo-catalytic methanolysis of PET to give DMT and EG in near quantitative yields. The organo-catalyst and solvent mixture can easily both be recirculated and also distilled to overcome the separation problems encountered using a metal catalyst (Scheme 1d).

The benign separation of solvent and catalyst enables reduction of the solvent to substrate ratio to 3:1.

The global warming potential (GWP) of the developed process was evaluated using LCA. We assessed the GWP of selective depolymerization PET in polycotton to fully recycle the cellulose fibers, producing DMT and EG, and compared it to the current PET treatment processes—including PET incineration and conventional PET recycling. The greener pathway of PET recycling is expected to be of interest to the chemical industry as it transitions toward lower carbon footprint solutions.

RESULTS AND DISCUSSION

Depolymerization of Polyesters. Our goal was to develop an environmentally sustainable methanolysis to give the DMT with high selectivity using a low solvent/PET ratio including any post-treatment. We initially ran the base-catalyzed depolymerization at $200\text{ }^\circ\text{C}$ using NEt_3 as catalyst as it is easily distilled, since the solvent recovery is a major factor for developing a sustainable procedure. It was found that running the depolymerization in MeOH and NEt_3 gave the desired DMT in 82% yield (Table 1, entry 1). This product crystallized and could be readily filtered as white crystals upon cooling. For our subsequent solvent recycling experiment, *vide infra*, our aim was to partially solubilize DMT, enabling us to introduce an additional batch of PET in the subsequent cycle. To achieve this, we explored various cosolvents that could effectively dissolve DMT. When the reaction was performed in methanol and toluene as a cosolvent, full conversion to DMT

Table 1. Optimization Table for the Depolymerization of PET Fibers^b

entry ^a	PET/solvent/Cat (v/v)	cosolvent	DMT (wt %)
1	1:0:0.13	none	82
2	1:1:0.13	toluene	88
3	1:1:0.13	xylene	86
4	1:1:0.13	anisole	66
5	1:1:0.13	cyclohexane	77
6	1:1:0.13	Me-cyclohexane	76
7	1:1:0.13	heptane	56
8	1:1:0.13	2-Me-THF	78
9	1:1:0.13	2-butanone	54
10	1:1:0.13	pinacolone	69
11	1:1:0.13	MEK	55
12	1:1:0.13	MTBE	40
13*	1:1:0.13	toluene	88
14	1:1:0	none	

^aReaction conditions: For all reactions, 0.3 g of PET was used at 200 °C for 2 h, and total volume of solvent used in each run was 4.5 mL.

^b*indicates the reaction performed in a Teflon reactor.

and EG was observed (Table 1, entry 2).¹⁹ A 1:1 ratio of toluene and methanol as a solvent mixture was used, and the reaction was run at 200 °C. Below this temperature, longer reaction times were required. When running the depolymerization at 130 °C, 10 h were required to obtain full conversion. Below 110 °C, no conversion was observed. It should be noted that 200 °C is a comparably low temperature for PET depolymerization where most technologies use temperatures between 200 and 300 °C. Subsequently, we screened several solvents with lower environmental impact, however with properties similar to toluene to determine if we could improve or maintain the yields of DMT and EG. However, we observed minimal differences in yield across these experiments (Table 1, entries 3–4). To improve the sustainability of the depolymerization, various environmentally benign solvents^{31,32} as compared to toluene such as cyclohexane, Me-cyclohexane, *n*-heptane, 2-MeTHF, 2-butanone, pinacolone, methyl ethyl ketone (MEK), and methyl *tert*-butyl ether (MTBE) were investigated to replace toluene (Table 1, entries 5–12). Unfortunately, none of these solvents was able to achieve the

same high yield as toluene. To rule out the possibility that trace amounts of metals from the reactor acted as catalysts in the reactions, the depolymerization was performed in a Teflon reactor giving the same results (Table 1, entry 13). The role of Net_3 as a catalyst was proven as no conversion was observed in the absence of Net_3 (Table 1, entry 14).

We characterized the products, DMT and EG, using ^1H NMR spectroscopy. The spectra were obtained in both CDCl_3 and D_2O . To analyze DMT, we dissolved the DMT crystals in CDCl_3 . Subsequently, we evaporated the filtrates containing solvents and confirmed the presence of EG by NMR in D_2O . The chemical shifts correspond to the literature values.¹³ The NMR analysis also disclosed that the products obtained are pure (>95%). The DMT sample was further analyzed by FT-IR where one clear carbonyl stretching frequency was detected at 1712 cm^{-1} . The reaction mixture was also subjected to gel permeation chromatography (GPC) and the Mn values clearly indicated that all the polymeric fibers degraded into the corresponding monomer (ESI). When the reaction was performed below 200 °C, oligomers were also observed in the GPC spectrum.

We hypothesized that the method could be applied to other polyesters and polycarbonates. Gratifyingly, we were able to fully depolymerize polycaprolactone (PCL), poly(lactic acid) (PLA) as shown in Table 2. Noteworthy, these reactions could be performed at 100 °C, running the reactions for 2 h. When the methodology was applied to poly(bisphenol A carbonate) (BPA-PC), a mixture of products was observed when the reactions were performed in methanol. However, when water was used as solvent, high selectivity to bisphenol A (BPA) was achieved. Thus, the methodology can successfully be applied to different polyesters.

Optimization of Mass- and Energy Balances. To reach a better environmental performance of the depolymerization with respect to the solvent and catalyst, we studied the possibility and efficiency to both recirculate and recycle the solvent and catalyst in the depolymerization of PET. The recirculation was performed by the addition of a new batch of PET (0.3 g) to the solvent mixture (3.95 g) comprising both DMT and EG after the reactions had been performed. We were able to recirculate the solvent system this way 5 times, giving a total PET: solvent ratio of 1:3 (1.5 g of PET: 3.95 g of

Table 2. Depolymerization of Different Polyesters^a

Entry	Polymer	Monomer	Yield (%)
1			95.0 ^a
2			>95.0 ^b
3			>95.0 ^c

^aReaction condition: (a) 0.3 g PCL, 4.5 mL (MeOH/ Net_3 , 93:7), 200 °C, 2 h. (b) 0.3 g PLA, 4.5 mL (MeOH/ Net_3 , 93:7), 100 °C, 2 h. (c) 0.3 g PBA, 4.5 mL (MeOH/ H_2O / Net_3 , 70:23:7), 160 °C, 2 h.

solvent). After these five runs, the solvent was recycled by distillation. During solvent recycling, PET degradation was monitored through GPC. In case of recirculating the solvent–catalyst mixture after first cycle, the PET fibers fully converted into DMT monomers, whereas in the fifth cycle, negligible amounts of oligomers were detected in the GPC (Supporting Information).

After performing the depolymerization for 5 cycles, it was possible to recycle >91% of the initial solvent/catalyst mixture. Thus, the solvent and EG loss was only 0.77 g for 1.5 g of PET. It should be noted that the methanol in the DMT is released upon repolymerization in the production of PET and can thus be fully recycled. Instead of adding more solvents or increasing the temperature in the distillation to recycle higher amounts of solvent and EG, the nonrecovered mixture can be used as process heat, *vide infra*. Thus, the overall mass balance for the depolymerization of PET is (eqs 1 and 2)

$$\text{input: solvent (3.45 g) + MeOH (0.50 g)}^* + \text{PET (1.5 g)} \quad (1)$$

$$\begin{aligned} \text{output: solvent (2.81 g) + DMT (1.44 g) + EG (0.43 g)} \\ \text{losses (0.77 g)} \end{aligned} \quad (2)$$

*Stoichiometric amount of MeOH to form DMT.

Life Cycle Assessment of Depolymerization of PET.

Life-cycle assessment (LCA), a method standardized by the ISO 14040 series, represents a comprehensive method to assess the environmental impact based on the emissions released throughout the life cycle—including the extraction or production of the background raw materials, their transport, and the final production or treatment process. In this study, LCA has been used to estimate the potential GWP of recycling one kg of PET fiber to DMT and EG and compare it to the existing PET waste treatment processes, including PET incineration and conventional PET recycling. The system boundary of the LCA is cradle-to-gate, which includes the production of input resources (*i.e.*, toluene, methanol, and triethylamine), transport, and the waste PET treatment process (Figure 1). The conventional “cut-off” approach is used to incorporate the impact of waste PET. The “cut-off” principle implies that the used PET product, considered waste, does not bear the environmental burden from the prior life/lives and thus carries zero impact within the presented recycling system.

PET recycling produces DMT, EG, and heat energy (steam energy) from the incineration of waste streams. These coproducts introduced on the market contribute to avoiding the production of DMT, EG, and heat through conventional sources. Hence, the system is credited for substituting the products (DMT, EG, and steam energy) in the market, thereby avoiding the environmental impact associated with producing them through conventional pathways, typically derived from fossil-based sources.

The study assesses the GWP using ReCiPe 2016 (Hierarchist) impact assessment methods, which use standardized characterization factors specified by the Intergovernmental Panel on Climate Change (IPCC). The GWP of the recycling process is 2.206 kg CO₂ equivalent, which includes the impact of raw material production, transportation, and the PET treatment process. The primary contributor to the GWP in this process is the flue gas emissions (mainly containing CO₂) during the PET treatment process. However, the process produces DMT, EG, and heat. The avoided burden from the

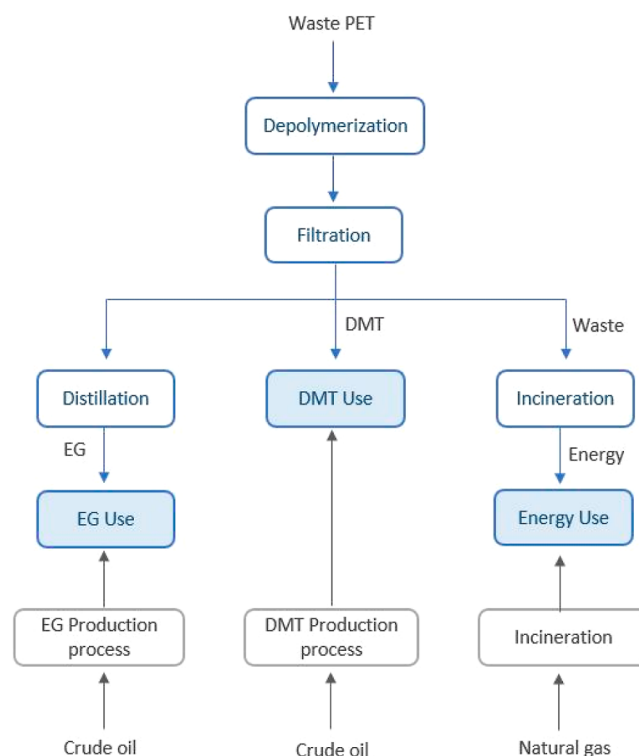


Figure 1. LCA system boundary of recycling PET fiber to DMT and EG. The gray boxes and flows represent the avoided production from the conventional fossil sources.

production of DMT, EG, and steam-energy production from fossil sources is 4.075 kg of CO₂ equivalent. As a result, recycling 1 kg of PET could potentially avoid the emission of 1.88 kg of CO₂ equivalent.

The GWP of incinerating an equal amount of PET exhibits a wide range, values ranging between −1.2 and 0.05 kg reported by Astrup,³³ 1.02 in the ELCD database, 1.4 reported by Chilton,³⁴ and 2.61 kg CO₂ equivalent reported by Aryan.³⁵ The GWP of existing recycling processes estimated in the literature varies from −0.7 to −1.5,³³ −1.64³⁶ and −1.457 and −1.7 kg CO₂ equivalent.³⁴ The existing recycling processes are environmentally better than incineration. However, the presented recycling has an even lower GWP than that of state-of-the-art PET recycling processes.

Application to Recover Both DMT and Cotton in a Polycotton Fabric. In textiles, PET is often used in a mixture with other fibers. Many clothes are composed of both cellulose fibers comprising cotton/viscose/lyocell and PET and this blend is termed polycotton.³⁷ Blended fabrics are even more challenging to recycle than fabrics consisting of only one material. To evaluate the applicability of the developed procedure, the methodology was tested to recycle PET in a polycotton fabric. The above conditions were applied to a postconsumer polycotton textile fabric sample. Dye-free polycotton (50% cotton and 50% PET) textile was used in the experiment. The reactions were performed using 0.300 g of polycotton fabric and 4.5 mL of MeOH/toluene/NEt₃ (50:50:7 v/v) solvent–catalyst mixture. The reaction was performed at 200 °C for 2 h where full depolymerization of PET into its monomers was observed. The solvent mixture comprising the DMT and EG were easily separated from the cotton through first filtration and the DMT was crystallized in 56% yield, and the solvents could be distilled off from the EG.

The cotton was isolated as a fluffy white pulp in 93% yield and with an intrinsic viscosity 472 mL/g and degree of polymerization of 655 (Figure 2). These values clearly indicate that the

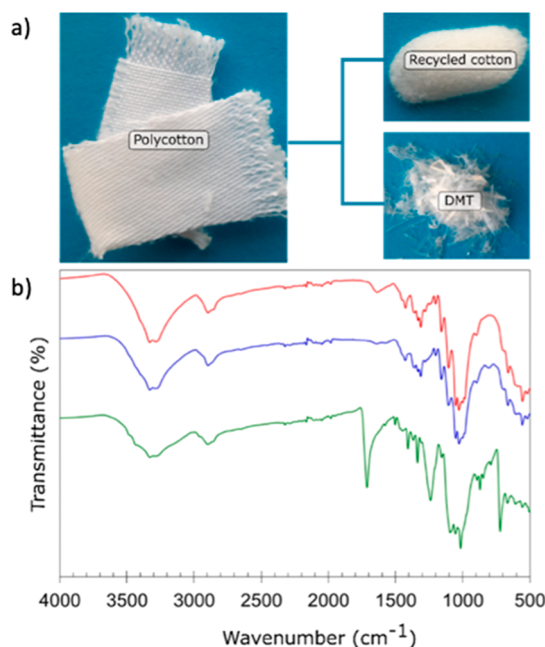


Figure 2. (a) Chemical recycling of cotton and chemical (DMT) from polycotton; (b) FT-IR spectra of starting polycotton (green), recycled cotton after chemical treatment (blue), and virgin cotton (red).

recycled cotton fiber is not degraded during the fractionation and the obtained pulp meets the specifications for both viscose and Lyocell production.^{17,38} We were able to scale up the reaction with polycotton to 20 g, giving similar results.

Mechanical Testing of Viscose Filaments from the Recycled Cotton. The cotton isolated during the depolymerization of PET was successfully used to prepare a viscose dope following a laboratory procedure originally described by Trieber using conditions similar to those for commercial dissolving pulp. The dope was further spun into viscose fibers.

The tensile properties of the individual filaments were measured in tensile mode by using an Instron 5944 mechanical testing instrument (Instron, U.S.A.) equipped with a 100 N load cell. The linear density (Table 3) of the individual

Table 3. Tensile Properties and Linear Density of 50 and 60% Viscose Filaments

	50% stretch	60% stretch
diameter (μm)	26.80 ± 4.21	19.79 ± 5.15
linear density (dtex)	9.19 ± 2.63	5.15 ± 3.82
tenacity (cN/tex)	7.47 ± 0.88	13.37 ± 1.53
elongation (%)	10.38 ± 3.27	18.38 ± 6.49

filaments was derived from the estimation of their average diameter in the optical microscope. For the 60% filaments, the diameter was decreased by 26% compared to the 50% filaments resulting in a linear density of 5.15 dtex which is higher but comparable to commercial viscose filaments (1.4 dtex).³⁹ The tenacity as well as the elongation (Table 3) of the 60% filaments was almost double compared to the tenacity of the 50% filaments, which confirms that stretching improves the

mechanical properties of individual viscose filaments. The tenacity of the 60% filaments (13.37 ± 1.53 cN/tex) is in the same range as other noncommercial cotton-derived viscose fibers (12.0–13.8 cN/tex)⁴⁰ and comparable to commercial viscose fibers (23.9 cN/tex).³⁹

CONCLUSIONS

Depolymerization of PET fibers into DMT and EG in excellent yields by using NEt_3 as organo-catalyst has been developed. The fiber depolymerization was performed at 200 °C for 2 h which is a comparably low temperature for PET depolymerizations. The use of NEt_3 as catalyst is crucial for the methanolysis depolymerization process of PET. By careful optimization of the solvent mixture, a high purity DMT was obtained through simple filtration and did not require any additional purification step. Previous methodologies have not reached high selectivity to monomers or used a catalyst that ends up in the product mixture where tedious postpurifications have been required. The solvent system used can first be recirculated without the requirement of energy demanding distillation up to 5 times; then simple filtration gives a pure DMT and distillation of a pure EG where 90% of the solvents can be recovered. The method was successfully applied to other polyesters: PLA, PCL, and PBA to afford the corresponding monomers. The LCA of PET depolymerization estimates a CO_2 footprint of the process to -1.88 kg CO_2 equivalent, lower than that of waste PET incineration or existing PET recycling technologies. In addition, we applied these reaction conditions on polycotton textile waste where the PET fibers were transformed into chemicals, and the cotton was isolated in pure form. The obtained cotton was further processed into dissolving grade pulp, from which a viscose dope was prepared and spun into viscose fibers of 50 and 60% stretch. The tenacity of the 60% stretched individual filaments was in the same range as nonwaste cotton-derived viscose filaments which paves the way for further use of the recycled cotton during the methanolysis in the production of textile cotton fibers. Thus, a benign recycling of polyesters to monomers combined with the use of the isolated cotton for producing viscose fibers has been developed, and the environmental sustainability of PET depolymerization has positively been evaluated by LCA. We hope that this study will inspire engineers to further develop this system for potential future commercialization.

EXPERIMENTAL SECTION

Experimental Procedure for PET Depolymerization. The PET fibers (0.300 g) were placed in a stainless-steel Swagelok reactor, and 4.5 mL of MeOH and 7% NEt_3 were added. The methanolysis reaction was performed at 200 °C for 2 h where full depolymerization of PET into its monomers was observed. After the reaction, the DMT spontaneously crystallized upon cooling down the reactor temperature. The obtained DMT crystals were isolated by simple filtration in 0.262 g (88% yield).

Solvent Recycling Experiment. The initial reaction was performed with 0.3 g of PET in the solvent mixture of 3.95 g of MeOH–toluene– NEt_3 (50/50/7 v/v). In the second cycle, the recirculation was performed by the addition of a new batch of PET (0.3 g/batch) to the product and solvent mixture. After five runs, the solvent was recycled by distillation. During the solvent recycling, the PET degradation was monitored through GPC.

Experimental Procedure for Chemical Recycling of Polycotton. The chemical recycling experiment was performed in a Swagelok metal reactor (7 mL capacity). 0.3 g of polycotton (50% cotton and 50% PET) textile fabric was placed in a reactor and treated with 4.5 mL of solvent–catalyst mixture MeOH–toluene– NEt_3 (50/

50/7 v/v) at 200 °C for 2 h. The PET in polycotton was depolymerized and formed DMT and EG. The obtained monomers DMT and EG were easily separated from the cotton through simple filtration. DMT (56%) was crystallized and the solvents could be distilled off from the EG. The remaining cotton was washed with the excess of methanol to remove degraded-PET monomers. Finally, a fluffy white pulp (cotton) was isolated in a 93% yield.

Experimental Procedure for Preparation of Viscose Dope and Fiber Spinning. Twenty-five grams of portion of the pulp was added to 800 g of 18% w/w in a stainless-steel vessel and was kept at 50 °C for 20 min to prepare alkali cellulose. The slurry was dewatered in a hydraulic press to a consistency corresponding to 32% cellulose in the sample. The alkalicellulose was then shredded mechanically to obtain smaller pieces for 35 min before being transferred into a plastic bottle. The bottle was thereafter placed in a water bath at 50 °C for depolymerization for 35 min. An amount of 42 g of alkalicellulose was then transferred to a glass reactor with a stirrer for sulphidation. The reactor was placed in a water bath at 28 °C for 2.5 h and CS₂ was added with a dosage of 40% w/w, based on cellulose. The cellulose xanthate formed was dissolved in sodium hydroxide, giving a final viscose dope a concentration of 9% cellulose and 6% sodium hydroxide. The dissolution was carried out in a dissolving vessel with stirring for 3 h at 7 °C, then for 16 h at 16 °C.

The wet-spinning of filaments was performed on an Aditya Birla Spinning pilot. The dope was extruded through a 40 hole spinneret (80 μm each) with an extrusion speed of 2.6 mL min⁻¹ into a coagulation bath containing 10 g/L of zink sulfate, 110 g/L of sulfuric acid (H₂SO₄), and 310 g/L of sodium sulfate (Na₂SO₄). The coagulation bath temperature was fixed at 48 °C. The filaments were stretched between godets of the spinning machine. The filaments were then collected and washed in a sequence using hot water (95 °C), hot dilute sulfuric acid (95 °C, 4.5 g/L), water (room temperature), warm dilute sodium hydroxide (60 °C, 0.4 g/L), and again water (room temperature). The filaments were air-dried at room temperature.

Experimental Procedure for Mechanical Testing of Individual Viscose Filaments. The individual viscose filaments were placed in a paper window with grip separation of 24 mm, and they were first placed in an optical microscope to estimate their average diameter. To minimize the error, 13 individual fibrils from each sample were measured in the optical and 6 measurements were performed on each individual filament having a sum of 78 diameter measurements per sample. The linear density in g km⁻¹ or tex was then calculated by multiplying the volume of the cylindrical filament with the theoretical density of viscose filaments (=1.52 g cm⁻³) divided by the length of the filaments. To estimate the tensile properties of the individual viscose filaments, the paper window was fixed between the grips in an Instron 5944 mechanical testing instrument (Instron, U.S.A.) equipped with a 100 N load cell. Once the paper window was placed between the grips, the side edges of the paper were cut to ensure that only the tensile properties of the individual filaments will be measured. The filaments tested were the same that were tested in the optical microscope, meaning that there were 13 filaments per sample for mechanical testing. The tenacity was calculated as the force at break divided by the linear density of the filaments, and the elongation was the tensile strain in % at break.

LCA

LCA was performed according to the ISO 14044 standard using four stages. System boundary of the study was cradle-to-gate; the conventional cutoff approach was used. Multi-functionality, energy of combustion, DMT, and EG were handled by system expansion. GaBi 8.7.0.18 software was used, and ecoinvent version 3.7.1 was used in the inventory. Climate change was chosen as impact category, by GWP₁₀₀, thus a 100 year's perspective was calculated using ReCiPe 2016 (Hierarchist) impact assessment.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.3c07435>.

Spectral data, and four stages of the LCA (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. S.M. and K.R.B. contributed equally.

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Notes

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